Practice workbook · Keeping a network secure: access rights, physical security and authentication

Why networks need protecting and the measures used: access rights, password policies, physical protection of servers, authentication methods and the difference between validation and authentication.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-07

1. Question 4 · part c · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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1. Question 4 · part c · 2 marks

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Worked-example retry · Attempt part c only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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1. Question 4 · part c · 2 marks · Working and answer

Working and answer

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2. Question 2 · part b · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

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Cloud storage provider

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(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	**Zafer**
Washing machine design drawings		
A folder of manuals for new machines		

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P 6 1 8 8 0 A 0 4 2 4

2. Question 2 · part b · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Zafer uses a browser to view pages on Pearson's website by typing a uniform resource locator into a browser. This is shown in **Figure 1**.



Figure 1

Pearson's machine	Zafer's machine
MAC address: 03-57-00-EC-4B-30	MAC address: 00-15-00-BC-9A-90
IPv4 address: 2.20.38.113	IPv4 address: 192.168.1.78
IPv4 Subnet Mask: 255.255.0.0	IPv4 Subnet Mask: 255.255.255.0
Website domain: www.pearson.com	Web browser
Main web page: https://www.pearson.com/uk/	Uniform resource locator: www.pearson.com

A domain name server is used in this process.

Identify the input to and the output from the domain name server.

(2)

Input

Output

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5

Turn over ▶

2. Question 2 · part b · 2 marks · Working and answer

Working and answer

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3. Question 2 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
- ☒ **B** Enforce a strong password policy
- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

(b) Danielle has received this email.

She is a customer of SafePayment.com but thinks this is a phishing attempt.

From: SafePayment.accounts@5afepayment.com
To: Danielle616
Subject: Account restricted (case SP-0011312-2021-06)

Dear Customer,

We have noticed some unusual activity on your account, so have stopped all payments.

Please use this link to log in and check your account.

Log in

Your normal log in will stoop working until you have done this.

Regards
SafePayment fraud prevention team.

(i) Danielle has circled the words **Dear Customer**.

Give **one** reason why the use of these words might make Danielle suspect that the email is a phishing attempt.

(1)

(ii) Draw circles around **two other** aspects of the email that might make Danielle suspect it is a phishing attempt.

(2)

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P 6 9 2 9 7 A 0 6 2 0

3. Question 2 · part a · 1 mark · Working and answer

Working and answer

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4. Question 1 · part b.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

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2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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4. Question 1 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

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2 .....

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(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

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(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

4. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part e · 2 marks

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Further practice · Attempt part e only.

4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

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5. Question 4 · part e · 2 marks

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Further practice · Attempt part e only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

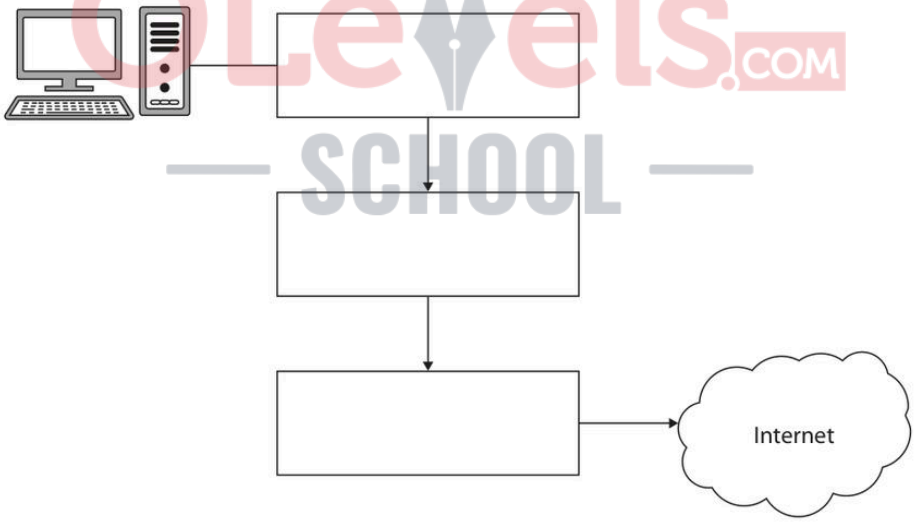
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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P 7 2 9 3 7 A 0 1 2 2 4

5. Question 4 · part e · 2 marks

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Further practice · Attempt part e only.

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(e) Users enter passwords when logging onto a network and when creating accounts using a web page.

Describe **one** difference between validation and authentication.

(2)

(Total for Question 4 = 12 marks)

Working and answer

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6. Question 1 · part b · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

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2 .....

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(b) Give **one** security risk of connecting computers in a network.

(1)

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6. Question 1 · part b · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b only.

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(c) **Figure 1** shows a network topology.

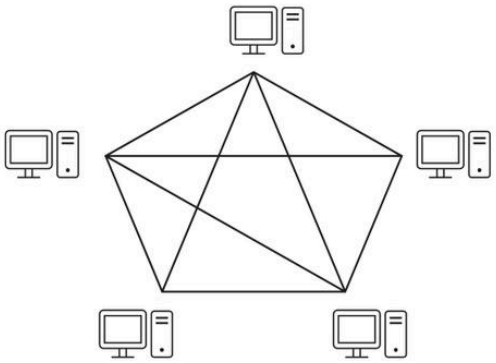


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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6. Question 1 · part b · 1 mark · Working and answer

Working and answer

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7. Question 4 · part a.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.ii only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

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2 .....

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(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

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2 .....

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7. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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